

Emily Kalinich
Professor Kinkhabwala
Computational Biology 373
May 6, 2026

Lophiiformes Neural Network

AI and organisms are complex, emerging, and changing fields. AI enables researchers to understand animal behavior better (Urbaniak et al., 2024). Setting up different machine learning programs can allow scientists to explore and analyze behavioral traits that were previously inaccessible due to human interference. Another area where AI has aided biology is neural networks, which can learn, detect patterns, and accurately predict (Norouzzadeh et al., 2018). Typically, the more data the network can learn from, the more accurate it will be. Neural networks have been used to predict and identify different animals (BIOCLIP) by detecting the distinct patterns (Stevens et al., 2024). Animals can be very distinct and also appear very similar. One order of organisms that appears vastly different is the order Lophiiformes. This order comprises five suborders, four of which are benthic: *Antennarioidei*, *Chaunacoidei*, *Lophioidei*, and *Ogcocephaloidei*; the other order, the bathypelagic *Ceratioidei* (Hart et al., 2022), lives in the deep sea. All of these classes live in very different environments; however, many of the species live in the deep sea (Monterey Bay Aquarium, n.d.). Due to the scarcity of prey, deep-sea fishes are assumed to be opportunistic feeders with generalized diets (Ebeling & Cailliet, 1974). Recent comparative studies have shown that deep-sea fishes exhibit greater morphological diversity than other fishes (e.g., slope or shelf fishes) (Miller et al., 2024). These studies have used jaw morphotypes to examine the diets of shallow-marine fish, demonstrating that, in a high-productivity environment, these fish exhibit a wide variety of morphologies (Binning et al., 2009). Jaw and tooth morphologies are crucial for understanding the ecological functions and the operation of feeding traits in specific deep-sea habitats. There will be a distinct pattern in head and jaw measurements across families, enabling the neural network to predict family membership accurately.

The anglerfish data used in this study were obtained from an online repository (Dyrad), specifically from the research paper “*Many ways to build an angler: diversity of feeding morphologies in a deep-sea evolutionary radiation*” by Heiple et al. 2023. These researchers used 57 museum specimen anglerfish and subjected them to microCT (Figure 1A). To measure the anglerfish, they are placed in this machine, and slice-by-slice images are taken to measure various variables. The researchers transformed the raw data to account for 3D features, since the CT scanner only measures in 2D. Once downloaded, there were 22 measurements; after accounting for time, the measurements were reduced to 13. The thirteen measurements are: avg.t, avg. t.u, avg.w, avg. w.u, avg.d, avg. d.u, gape, premax, lower.jaw, tooth.reg, desc.proc, tooth.count, meanspace. Two different groupings with varying numbers of categories were created to compare the neural network’s accuracy. The data were first grouped by creating histograms for each measurement and marking which species were in each histogram bin (Figure 1B). The color-coded graph yielded five groups at the end, using all the specimens (Figure 1C). The other

training loss and shows less improvement in error. In the second group, the training accuracy was only ~30% and showed a stair-like pattern (Figure 2C). The training-validation appears very erratic, and formed three “steps.” These two lines are showing a pattern of underfitting. Finally, looking at the training loss from group two compared to the first group, it starts much higher at 2.7 rather than ~1.7; it also shows a similar pattern of a consistent decrease; however, it ends right after the first group starts at ~1.4 (Figure 2D). The validation loss converges later than the training loss, and also shows a greater decrease in the validation loss. Both of the groups showed interesting accuracy, with the lower number categories (5) having some overfitting and the higher number of categories (9) having underfitting.

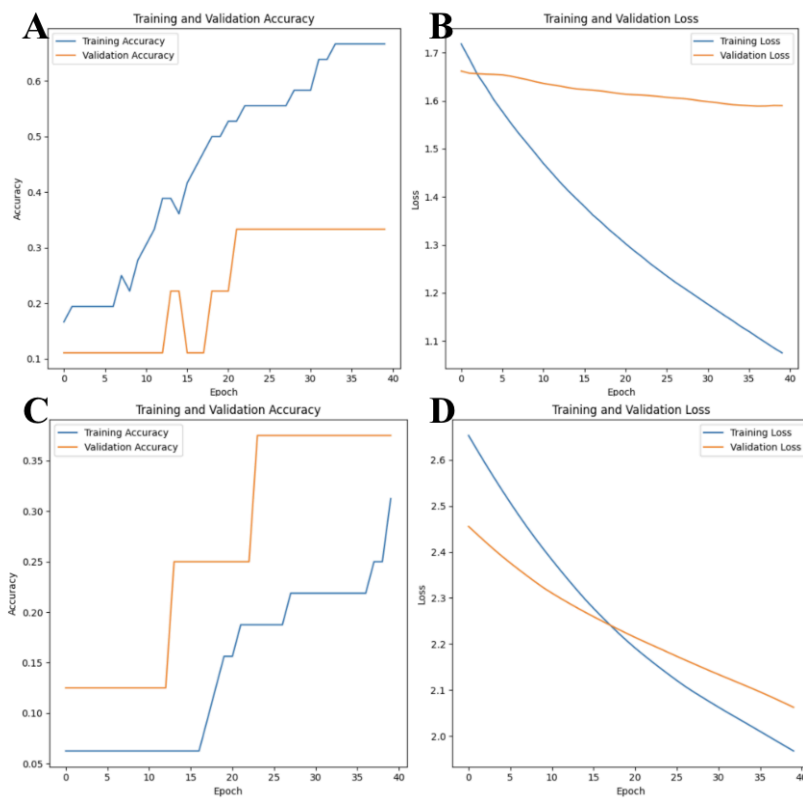


Figure 2. Training and validation accuracy and loss for both groups. A) 5 different groups showing a training accuracy of ~70% and validation accuracy of ~35%. B) 5 different groups showing the training loss starting at ~1.7 and ending around ~1.3, and the validation loss starting at ~1.7 and ending around ~1.6. C) 9 different groups showing the training accuracy ending at ~30% and the validation accuracy ending at ~37%. D) 8 different groups showing the training loss starting at ~2.65 and ending at ~2.0 and the validation loss starting at

~2.5 and ending at ~2.3.

A confusion matrix is an easy-to-read representation of the accuracy for each specific group, showing how well the neural network performed within each group. A perfect confusion matrix shows the highest value along the diagonal from the top left to the bottom right; this indicates a neural network that accurately predicted every test specimen. The first group's confusion matrix shows that the neural network performed very well on group 0 (Figure 3A). Looking at groups 1, 2, the network did not do too well, mispredicting three times in group 2 and mispredicting twice in group 1. Finally looking at groups 3 and 4 it showed a somewhat

accurate prediction with the values being 2. The second group, tasked with predicting 9 categories, no category was predicted accurately, the highest accuracy is for predicting group four in group one (Figure 3B).

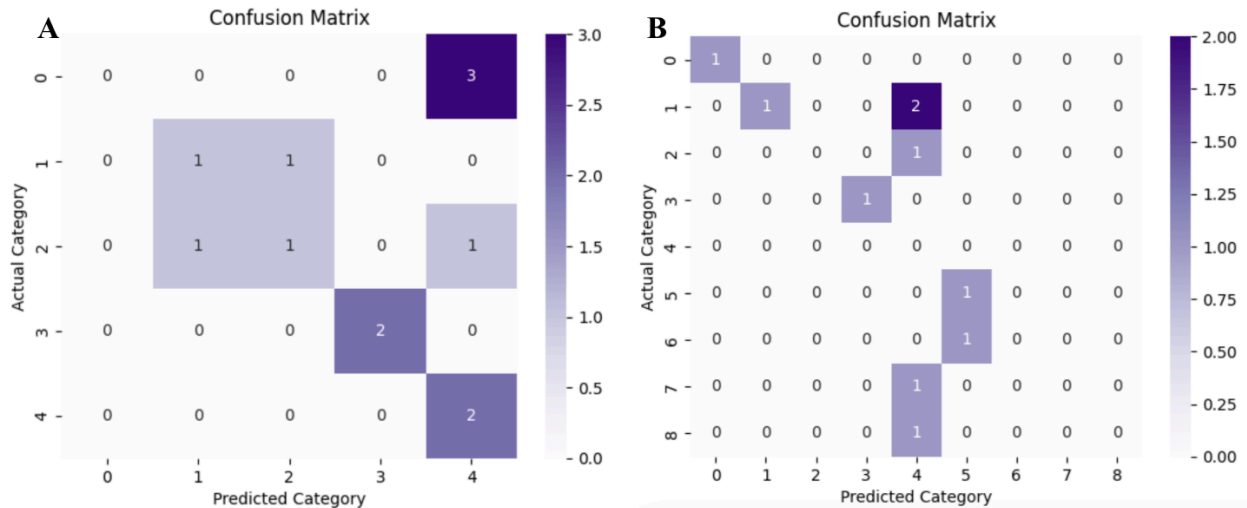


Figure 3. Confusion matrix of both groups. A) With 5 different categories the network accurately predicted group 0 with a value of 3, some mistakes in groups 3 and 4 with a value of 2, groups 1, 2, and 3 were not accurately predicted with a value of 1. B) With 8 different categories the network was 50% accurate on groups 0, 1, 3, and 5 with a value of 1, and was not able to predict groups 2, 4, 6, 7, and 8.

Overall, the neural networks were unable to predict the different anglerfish groups accurately. The first group, organized by histogram generalizations, showed that the neural network was overfitting and only learned the measurements of the given species rather than the overall patterns. This means the data groups were too complex, and the network learned the noise, resulting in high accuracy on the training data but lower accuracy on new data (Sahoo, 2024). In the other group, the model was underfitting, meaning the model was too simple and unable to learn general patterns, resulting in poor performance on both the training and test data. Although group 1 showed overfitting, the confusion matrix indicated that it still performed well on new data. It accurately predicted group 0 and somewhat accurately predicted groups 1, 3, and 4, but quite poorly on group 2. At the same time, the other groups' confusion matrix showed an underfitting model: the network really only understood the pattern for group 1 and tried to predict almost every other group as group 1. In the future, this experiment should be redone by reprocessing the groups. From this information, knowing that the first group was too complex and the second group was too simple, finding an intermediate group will provide the network with the best possible data to detect a pattern without over- or under-fitting. Another factor that will improve the model's accuracy is increasing the sample size. It is difficult to collect lophiiform samples; however, the greater the number, the more likely the model will be accurate. If this neural network was to work it would greatly improve the field of marine biology, helping with potentially introducing new or unknown species or figuring out other patterns that maybe weren't visible before.

AI Statement: No AI was used to create this paper, grammarly was used to help with punctuation and grammar.

Resources:

- Binning, S. A., Chapman, L. J., & Cosandey-Godin, A. (2009). Specialized morphology for a generalist diet: Evidence for Liem's Paradox in a cichlid fish. *Journal of Fish Biology*, 75(7), 1683–1699. <https://doi.org/10.1111/j.1095-8649.2009.02421.x>
- Ebeling, A. W., & Cailliet, G. M. (1974). Mouth size and predator strategy of midwater fishes. *Deep Sea Research A*, 21, 959–968. [https://doi.org/10.1016/0011-7471\(74\)90028-X](https://doi.org/10.1016/0011-7471(74)90028-X)
- Hart, P. B., Arnold, R. J., Alda, F., Kenaley, C. P., Pietsch, T. W., Hutchinson, D., & Chakrabarty, P. (2022). Evolutionary relationships of anglerfishes (Lophiiformes) reconstructed using ultraconserved elements. *Molecular Phylogenetics and Evolution*, 171, 107459. <https://doi.org/10.1016/j.ympev.2022.107459>
- Miller, E. C., Faucher, R., Hart, P. B., Rincón-Sandoval, M., Santaquiteria, A., White, W. T., Baldwin, C. C., Miya, M., Betancur-R, R., Tornabene, L., Evans, K., & Arcila, D. (2024). Reduced evolutionary constraint accompanies ongoing radiation in deep-sea anglerfishes. *Nature Ecology & Evolution*, 9(3), 474–490. <https://doi.org/10.1038/s41559-024-02586-3>
- Monterey Bay Aquarium. (n.d.). *Deep sea anglerfish*. Retrieved May 5, 2026, from <https://www.montereybayaquarium.org/animals/animals-a-to-z/deep-sea-anglerfish>
- Norouzzadeh, M. S., Nguyen, A., Kosmala, M., Swanson, A., Palmer, M. S., Packer, C., & Clune, J. (2018). Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning. *Proceedings of the National Academy of Sciences of the United States of America*, 115(25), E5716–E5725. <https://doi.org/10.1073/pnas.1719367115>
- Sahoo, P. (2024). *Machine Learning vs Neural Networks*. GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/machine-learning-vs-neural-networks/>
- Stevens, S., Wu, J., Thompson, M. J., Campolongo, E. G., Song, C. H., Carlyn, D. E., Dong, L., Dahdul, W. M., Stewart, C., Berger-Wolf, T., Chao, W.-L., & Su, Y. (2024). *BioCLIP: A Vision Foundation Model for the Tree of Life* (arXiv:2311.18803). arXiv. <https://doi.org/10.48550/arXiv.2311.18803>
- Urbaniak, R., Xie, M., & Mackevicius, E. (2024). Linking cognitive strategy, neural mechanism, and movement statistics in group foraging behaviors. *Scientific Reports*, 14(1), 21770. <https://doi.org/10.1038/s41598-024-71931-0>